

PRODUCT REVIEW SENTIMENT CLASSIFICATION FOR E-COMMERCE PLATFORMS

Problem Definition

E-commerce platforms face significant challenges in efficiently analyzing multilingual product reviews for sentiment. Existing solutions often demand substantial hardware or outsourcing, limiting accessibility and scalability for businesses seeking to understand customer feedback

Overview

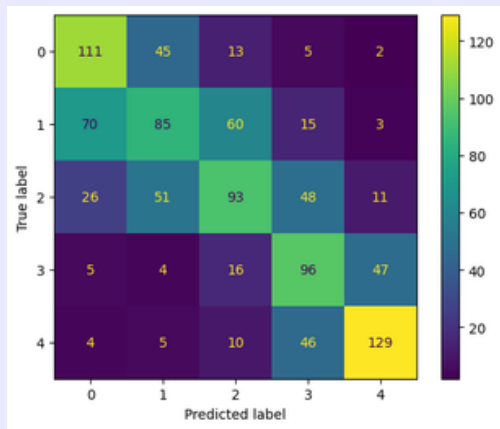
This project introduces an accessible AI model for multilingual product review sentiment classification, designed to operate without the need for professional-grade computational resources.

Project Goals

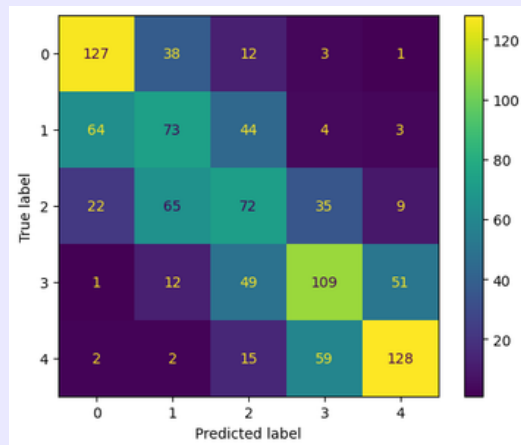
This project is about the creation of an AI model that enables its users to perform sentiment classification in a wide range of languages on written product reviews without the need for professional-grade or outsourced hardware/computational resources. This is done by combining two AI models: LibreTranslate and ModernBERT. LibreTranslate is used for identifying the language in which reviews are written, and translating the reviews into English if they're in a different language, and ModernBERT is used for classifying the sentiment of English reviews.

Evaluation

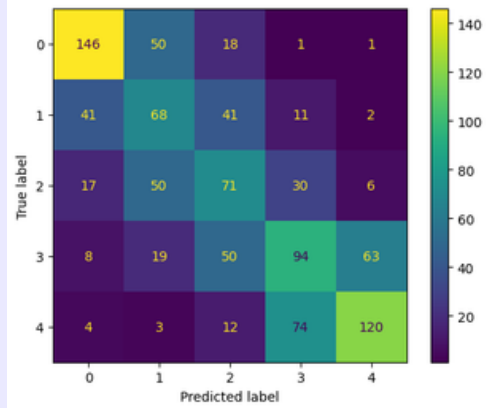
German language confusion matrix. acc 51.4%



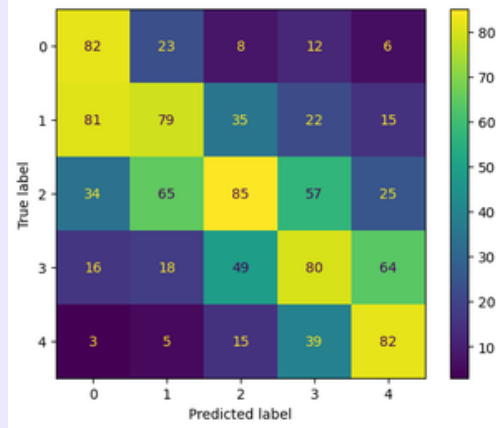
Spanish language confusion matrix. acc 50.9%



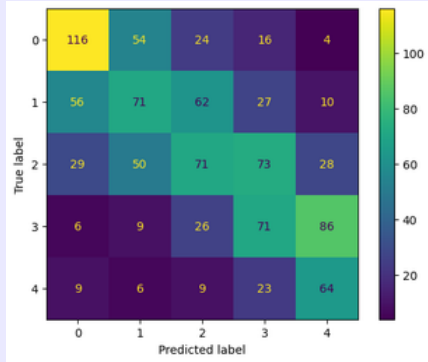
French language confusion matrix. acc 49.9%



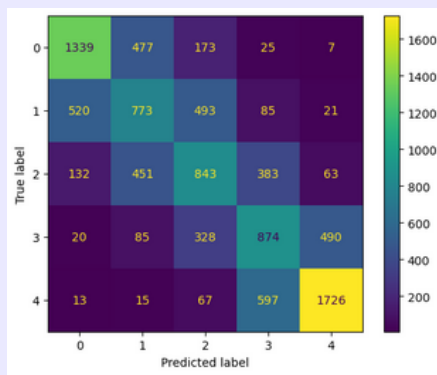
Japanese language confusion matrix. acc 40.8%



Chinese language confusion matrix. acc 39.3%



English language confusion matrix. acc 55.55%



Dataset

The dataset we used is called "Amazon Reviews Multi" (<https://www.kaggle.com/datasets/mexwell/amazon-reviews-multi>). Label-wise, it is a perfectly balanced dataset. It contains 210000 English language product reviews and, alongside the review body, has features: product category, and review title.

Conclusion

This model gave us an English language accuracy of 55.55%. For comparison, a randomly guessing classifier would have an accuracy of 20%. We found that the ratio of dataset sizes for the three training stages didn't have a large effect on model accuracy, all resulting in accuracies around 55% on English reviews. Although the model's accuracy on 5-star sentiment classification may not seem too impressive, its accuracy on 3-label sentiment classification is quite respectable, at 76% in English reviews. For comparison, Professional models achieve around 80% to 90%.

Future steps

The obvious improvements for this model are to train for longer on more data and more long reviews, and to utilize higher quality translation software than LibreTranslate. We could also train on pre-translated reviews from different languages.